Online Resource 1

Recoverability-aware records for spatial relations under incomplete reference-frame metadata

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S1 Formal assumptions and degeneracies

The relational path assumes labelled Euclidean chord distances of embedding dimension at most three and known metric scale. Sparse graphs may admit flexes or multiple discrete realizations. Repeated points, collinear or coplanar triples, axis-pole locations and projected antipodes are degeneracies and are reported as such rather than forced into a directional classification. A directed axis is required for axial ordering; signed relative longitude requires a physically recoverable chiral witness; absolute longitude additionally requires a directed meridian. A stored orientation bit without independent physical meaning does not remove the reflection ambiguity.

S2 Decoder algorithm

Step 1. Validate structure and semantics; a fatal issue returns invalid_record for every task, and component issues disable the affected anchor or path.

Step 2. Coordinate path: available only if the reference-system block is complete, the axis matrix is orthonormal and the coordinate hash verifies. Coordinates are mapped to the declared conventional frame and propagated with velocities when the query epoch differs from the coordinate epoch.

Step 3. Relational path: complete distances, augmented by origin radii when an independent origin anchor exists, are embedded by classical multidimensional scaling using the three largest positive eigenpairs of the double-centred squared-distance matrix. Valid axis anchors fix the axis by least squares, a valid physical chirality witness selects the reflection, and a meridian anchor fixes the rotation. Missing anchors leave the corresponding residual group.

Step 4. Cross-checks: coordinate-derived distances are compared with relations at the stated coverage factor; coordinate-derived latitudes and chirality are compared with independent anchors. Disagreement withdraws the coordinate path for orientation tasks and returns distance answers as set-valued with both candidates.

Step 5. Per-task state: unique when one consistent value is available; unique_up_to_group with the group name when an anchor is missing; probabilistic when the north–south sign probability under the axis-anchor uncertainty is below 0.975; non_identifiable when no path supports the task.

Proper and reflection-permitting Procrustes alignment for the geometric checks use a singular-value decomposition with an explicit determinant constraint. Rigidity uses the numerical rank of the three-dimensional rigidity matrix with an SVD tolerance scaled by matrix dimensions and machine precision.

S3 Schema and examples

The JSON Schema (draft 2020-12 vocabulary) requires the top-level fields schema, schema_version, record_id, entities. The complete schema, intact examples for representations A, B, BR, C, D and E, and damaged examples for scenarios S4, S9 and S10, the complete numerical registry and the benchmark outputs are provided in Online Resource 2 (ESM_2.zip).

S4 Configuration

```
seed: 20260924
synthetic:
  n_points: 24
  n_frame_trials: 250
  monte_carlo_trials: 4000
  ellipsoid:
    name: WGS84
    semi_major_axis_m: 6378137.0
    inverse_flattening: 298.257223563
  ellipsoid_sensitivity:
    name: GRS80
    semi_major_axis_m: 6378137.0
    inverse_flattening: 298.257222101
  anchor_corruption_probabilities: [0.0, 0.01, 0.05, 0.1, 0.2, 0.4]
  angular_noise_degrees: [0.0, 0.01, 0.05, 0.1, 0.25, 0.5]
rigidity:
  n_points: [6, 8, 10, 12, 16]
  edge_probabilities: [0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0]
  trials: 2000
  seed_offset: 1000
real_data:
  source: data/raw/itrf2020/ITRF2020_SLR.SSC.txt
  reference_epoch: 2015.0
  comparison_epoch: 2025.0
  max_coordinate_sigma_m: 0.02
  n_sites: 16
records:
  reference_epoch: 2015.0
  comparison_epoch: 2025.0
  sparse_edge_probability: 0.5
  perturbed_sites: [1, 5, 9]
  perturbation_m: 0.05
  missingness_rates: [0.05, 0.1, 0.2, 0.3, 0.5]
  missingness_conventions: [standard, left_handed, plate_fixed_rotated]
  missingness_replicates: 200
  calibration_axis_noise_radians: [0.01, 0.03, 0.1]
  calibration_replicates: 200
  timing_repeats: 50
  seed_offset: 3000
```

S5 ITRF2020 selected stations

Table S1 Selected ITRF2020 SLR stations and epochs.

Domes	Site name	Technique id	Latitude degrees	Longitude degrees	Reference epoch	Comparison epoch
40445M004	MAUI	7119	20.579588011218565	-156.25692679418262	2015.0	2025.0
40497M001	Monument Peak	7110	32.716542819713	-116.4226784499638	2015.0	2025.0
40442M006	FORT DAVIS	7080	30.5117098111711	-104.01519959418388	2015.0	2025.0
40451M105	Washington	7105	38.83248993117698	-76.8276968652703	2015.0	2025.0
48081S001	Campus Universi	7407	-15.67269811883136	-47.86529338063169	2015.0	2025.0
13402S007	SAN FERNANDO	7824	36.28149367442857	-6.2053067139107805	2015.0	2025.0
14001S007	Zimmerwald	7810	46.68520770658857	7.465224516900016	2015.0	2025.0
30302M003	Hartebeesthoek	7501	-25.73887838935569	27.6861772862697	2015.0	2025.0
25603S001	Baikonur Cosmod	1887	45.51259179864986	63.34241682566955	2015.0	2025.0
12372S001	Altay	1879	51.15604735269942	82.17729836577405	2015.0	2025.0
12338S004	Badary	1890	51.58283480322939	102.23536268348586	2015.0	2025.0
50107M001	Yarragadee	7090	-28.883434983501672	115.346753302934	2015.0	2025.0
50119S003	Mount Stromlo	7825	-35.13482884193121	149.00988228866052	2015.0	2025.0
12734S008	Matera	7941	40.4585779422104	16.704614547110 5	2015.0	2025.0
13212S001	Herstmonceux Ca	7840	50.67885528360768	0.3361285649584317	2015.0	2025.0
10002S002	Grasse (OCA)	7845	43.56246101585594	6.9215764612823 78	2015.0	2025.0

S6 Ellipsoid sensitivity and reference-frame taxonomy

Table S2 WGS84–GRS80 sensitivity of chord distances and directional classifications.

Metric	Value	Unit
ellipsoid_maximum_coordinate_difference_m	0.0001210656051966	m
ellipsoid_median_coordinate_difference_m	0.0001025915363387	m
ellipsoid_maximum_chord_distance_difference_m	0.0002250224351882	m
ellipsoid_median_chord_distance_difference_m	4.020938649773598e-05	m
ellipsoid_axial_ordering_agreement	1.0	proportion
ellipsoid_signed_longitude_agreement	1.0	proportion

Table S3 Reference-frame classes, physical anchors and principal failure modes.

Frame	Physical anchor	Preserving transformations	Temporal durability	Principal failure mode	North south support	East west support
Terrestrial reference frame	realized station coordinates and velocities	time-dependent Euclidean/Helmert mapping	high for modern epochs	network/model discontinuity	yes	yes
Celestial reference frame	extragalactic source directions	celestial rotations	high	source structure and realization change	with Earth orientation	with Earth orientation
Spin-axis frame	directed angular-momentum/rotation axis	rotation about axis and axial reflection	medium	polar motion and true polar wander	yes	only with physical chirality
Plate-fixed frame	rigid plate model	plate Euler rotation	medium	deformation and plate reorganization	model-dependent	model-dependent
Paleomagnetic frame	remanent field direction under field assumptions	axial rotations	geological	remagnetization and GAD violations	paleolatitude	not from paleomagnetism alone
Hotspot/mantle frame	hotspot tracks or mantle structures	model-specific plate rotations	geological	mantle motion and sparse anchors	model-dependent	model-dependent
Relative tectonic network	distances/ rotations among blocks	network gauge transformations	geological	topology loss and deformation	only with axial anchor	only with orientation anchor
Hybrid redundant frame	independent geometric, axial, and physical anchors	intersection of preserved symmetries	highest conditional robustness	correlated anchor failure	yes	yes

S7 Supporting figures

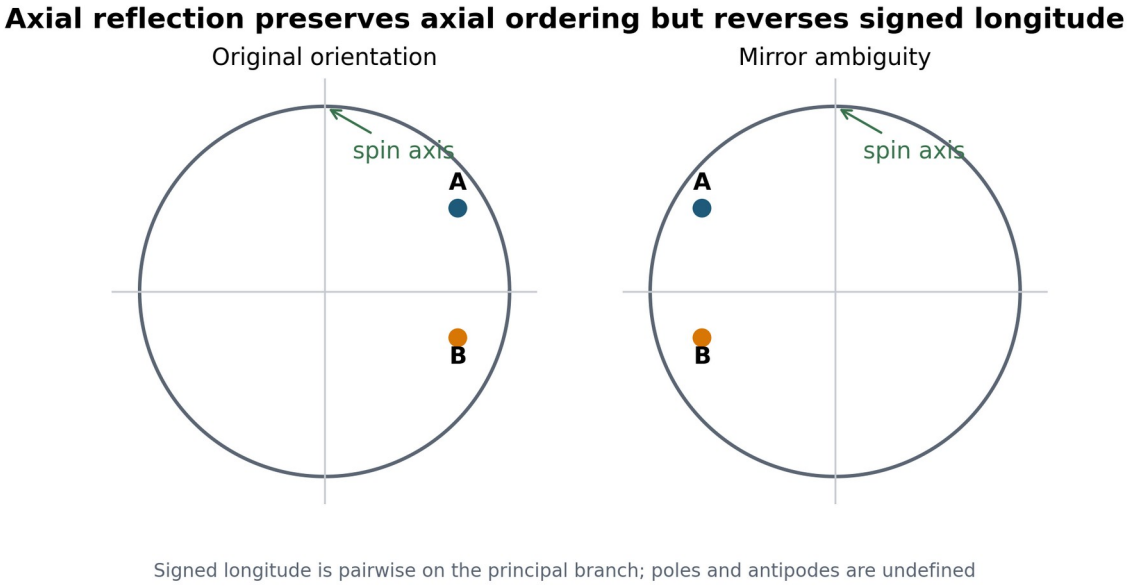


Fig. S1 Direction asymmetry: proper rotations preserve north–south and east–west relations; an axial reflection preserves north–south and reverses east–west.

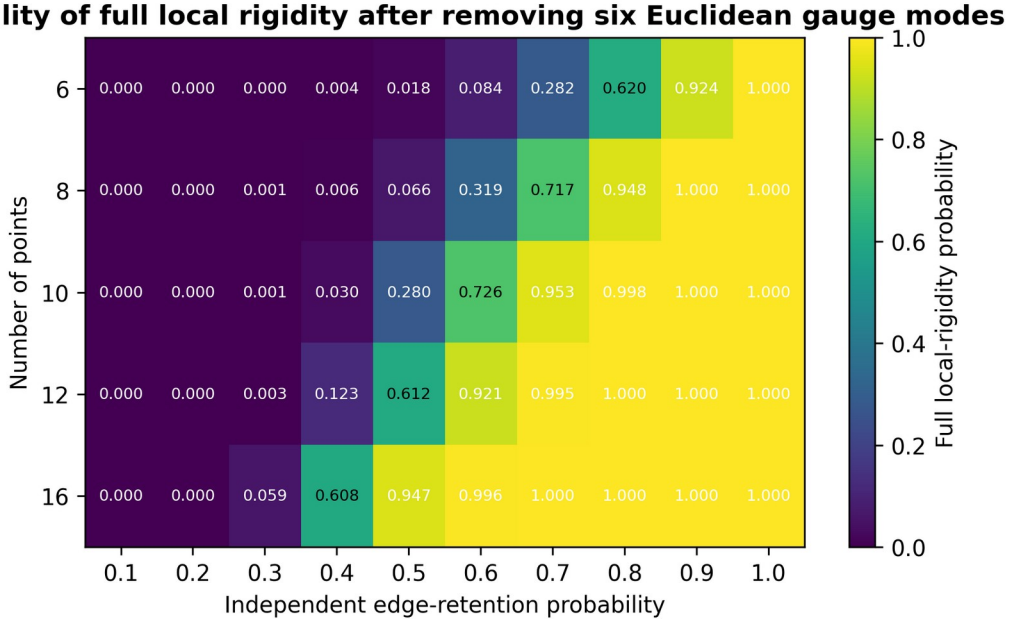


Fig. S2 Local-rigidity phase diagram for generic random three-dimensional frameworks.

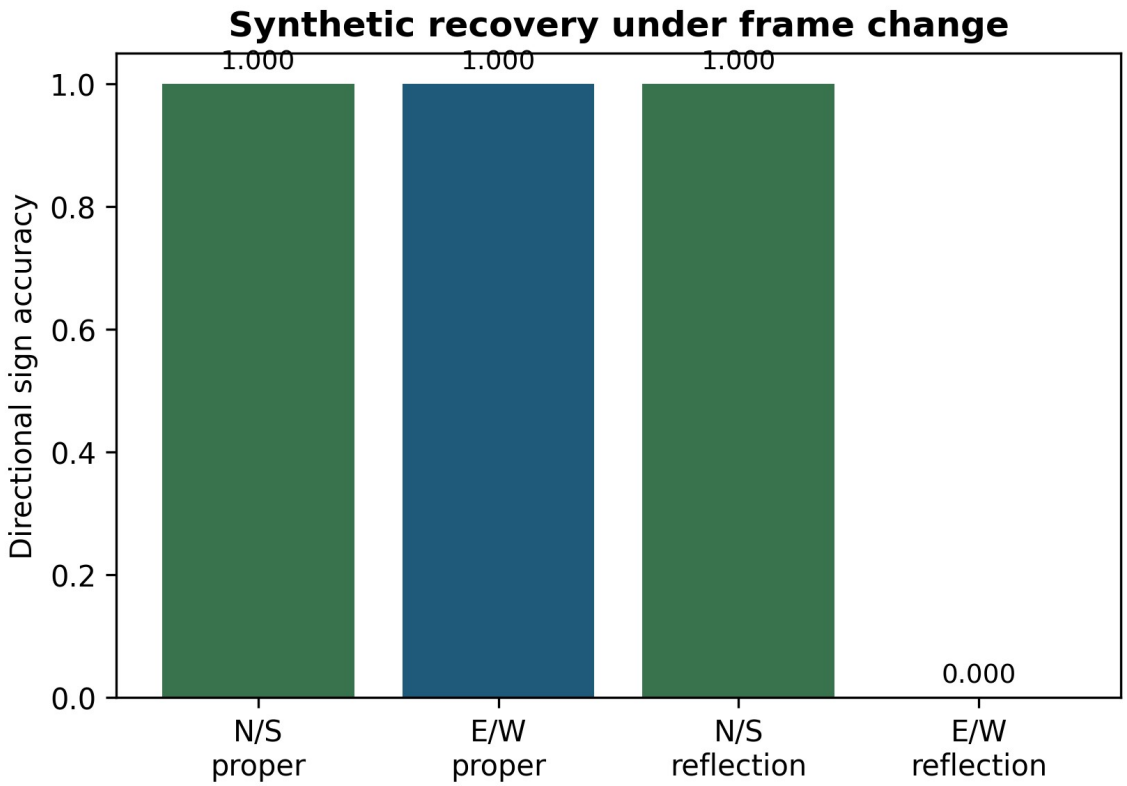


Fig. S3 Synthetic benchmark of complete-distance reconstruction with proper-only and reflection-permitting alignment.

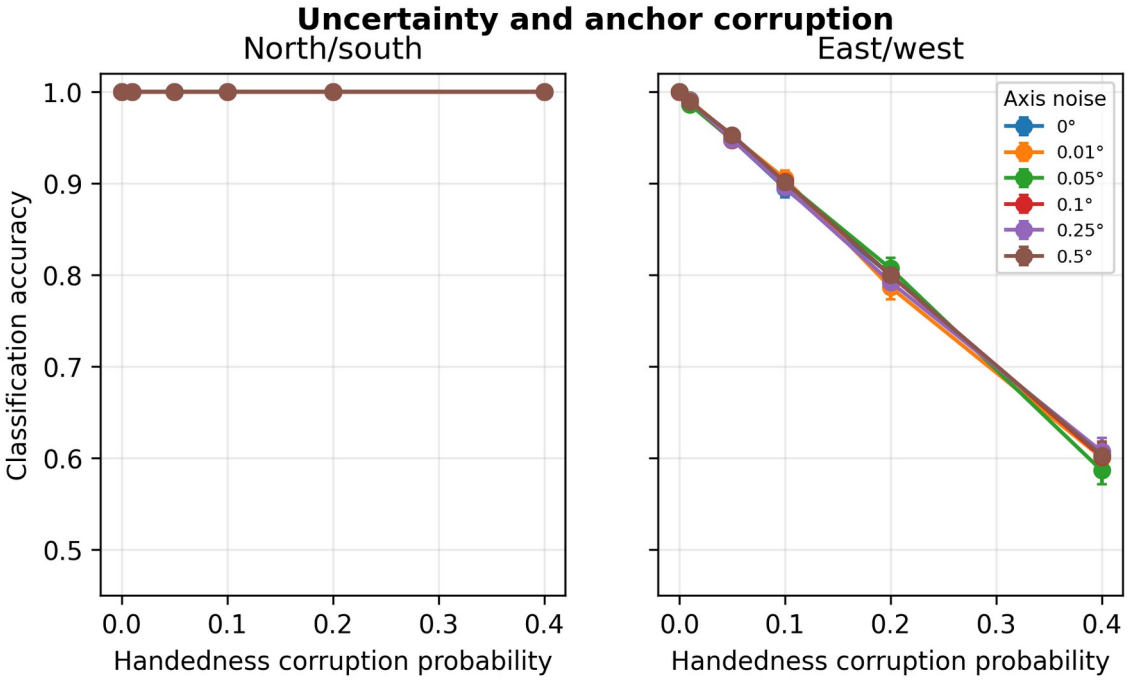


Fig. S4 Seeded axis-noise and orientation-corruption sensitivity of directional relations.

S8 Numerical registry

Table S4 Machine-generated values cited in the manuscript. The complete registry of 976 values is results/manuscript_values.csv in Online Resource 2.

Metric	Value	Source	Unit
synthetic_number_of_points	24.0	synthetic_benchmark	proportion_or_m
synthetic_frame_trials	250.0	synthetic_benchmark	proportion_or_m
synthetic_monte_carlo_trials_per_condition	4000.0	synthetic_benchmark	proportion_or_m
distance_reconstruction_rmse_allow_reflection_m	4.0247437518683874e-09	synthetic_benchmark	proportion_or_m
distance_reconstruction_rmse_proper_only_m	6136173.251345327	synthetic_benchmark	proportion_or_m
proper_rotation_north_south_accuracy	1.0	synthetic_benchmark	proportion_or_m
proper_rotation_east_west_accuracy	1.0	synthetic_benchmark	proportion_or_m
reflection_north_south_accuracy	1.0	synthetic_benchmark	proportion_or_m
reflection_east_west_accuracy	0.0	synthetic_benchmark	proportion_or_m
n_selected_sites	16.0	itrf2020_demonstration	count
maximum_coordinate_standard_uncertainty_m	0.02	itrf2020_demonstration	m
distance_reconstruction_rmse_m	3.236372328775209e-09	itrf2020_demonstration	m
axial_reflection_north_south_accuracy	1.0	itrf2020_demonstration	proportion
axial_reflection_east_west_accuracy	0.0	itrf2020_demonstration	proportion
maximum_pairwise_distance_change_m	0.6906252605840564	itrf2020_demonstration	m
median_pairwise_distance_change_m	0.0907241012901067	itrf2020_demonstration	m
reference_epoch	2015.0	itrf2020_demonstration	decimal_year
comparison_epoch	2025.0	itrf2020_demonstration	decimal_year
random_seed	20260924.0	configuration	integer
ellipsoid_maximum_coordinate_difference_m	0.0001210656051966	ellipsoid_sensitivity	m
ellipsoid_median_coordinate_difference_m	0.0001025915363387	ellipsoid_sensitivity	m
ellipsoid_maximum_chord_distance_difference_m	0.0002250224351882	ellipsoid_sensitivity	m

Metric	Value	Source	Unit
ellipsoid_median_chord_distance_difference_m	4.020938649773598e-05	ellipsoid_sensitivity	m
ellipsoid_axial_ordering_agreement	1.0	ellipsoid_sensitivity	proportion
ellipsoid_signed_longitude_agreement	1.0	ellipsoid_sensitivity	proportion
rigidity_largest_network_n_points	16.0	rigidity_phase_diagram	count
rigidity_trials_per_cell	2000.0	rigidity_phase_diagram	count
rigidity_95pct_threshold_edge_probability	0.6	rigidity_phase_diagram	probability
rigidity_probability_at_95pct_threshold	0.996	rigidity_phase_diagram	proportion
rigidity_95pct_threshold_95ci_lower	0.992126535703962	rigidity_phase_diagram	proportion
rigidity_95pct_threshold_95ci_upper	0.9979717533930348	rigidity_phase_diagram	proportion
rigidity_edge_probability_before_95pct_threshold	0.5	rigidity_phase_diagram	probability
rigidity_probability_before_95pct_threshold	0.9465	rigidity_phase_diagram	proportion
rigidity_probability_before_95pct_threshold_95ci_lower	0.935754264314357	rigidity_phase_diagram	proportion
rigidity_probability_before_95pct_threshold_95ci_upper	0.955533812463484	rigidity_phase_diagram	proportion
north_south_accuracy_at_0.1_chirality_corruption	1.0	uncertainty_benchmark	proportion
east_west_accuracy_at_0.1_chirality_corruption	0.895	uncertainty_benchmark	proportion
east_west_accuracy_at_0.1_chirality_corruption_95ci_lower	0.8851180041126476	uncertainty_benchmark	proportion
east_west_accuracy_at_0.1_chirality_corruption_95ci_upper	0.9041240356884884	uncertainty_benchmark	proportion
north_south_accuracy_at_0.4_chirality_corruption	1.0	uncertainty_benchmark	proportion
east_west_accuracy_at_0.4_chirality_corruption	0.60225	uncertainty_benchmark	proportion
east_west_accuracy_at_0.4_chirality_corruption_95ci_lower	0.5869914415518415	uncertainty_benchmark	proportion
east_west_accuracy_at_0.4_chirality_corruption_95ci_upper	0.6173123522954146	uncertainty_benchmark	proportion
rec_field_mcar_r0p05_B_records_with_false_certainty	0.0183333333333333	record_benchmark	proportion

Recoverability-aware spatial-relation records

Metric	Value	Source	Unit
rec_field_mcar_r0p1_B_records_with_false_certainty	0.015	record_benchmark	proportion
rec_field_mcar_r0p2_B_records_with_false_certainty	0.0383333333333333	record_benchmark	proportion
rec_field_mcar_r0p3_B_records_with_false_certainty	0.0566666666666666	record_benchmark	proportion
rec_field_mcar_r0p5_B_records_with_false_certainty	0.0333333333333333	record_benchmark	proportion
rec_ablation_full_false_certainty	0.0	record_benchmark	proportion
rec_ablation_no_relations_false_certainty	0.0345262096774193	record_benchmark	proportion
rec_ablation_no_chirality_anchor_false_certainty	0.0171370967741935	record_benchmark	proportion
rec_ablation_chirality_digital_copy_false_certainty	0.0171370967741935	record_benchmark	proportion
rec_calibration_noise0p01_brier_score	0.0048921897422517	record_benchmark	proportion
rec_calibration_noise0p03_brier_score	0.016845723745199	record_benchmark	proportion
rec_calibration_noise0p1_unique_error_rate	0.0028278543655001	record_benchmark	proportion
rec_calibration_noise0p1_brier_score	0.0665927396313885	record_benchmark	proportion
rec_n_items_per_scenario	496.0	record_benchmark	count
rec_n_scenarios	16.0	record_benchmark	count
rec_missingness_replicates	200.0	record_benchmark	count
rec_distance_tolerance_m	0.01	record_benchmark	m
rec_longitude_tolerance_degrees	1e-05	record_benchmark	degree
rec_perturbation_m	0.05	record_benchmark	m
rec_sparse_edge_probability	0.5	record_benchmark	proportion
rec_seed	20263924.0	record_benchmark	integer

S9 Test coverage

The automated suite covers rigid-transform distance invariance, proper and reflection-permitting alignment, simplex chirality, local ENU orthonormality, direction invariance and asymmetry, branch wrapping, pole and antipode degeneracies, rigidity rank, ITRF parsing, seeded reproducibility, schema validity, round-trip serialization, all six representations, group-valued and set-valued states, invalid-witness detection, digital-copy negative control and registration of every benchmark value.